

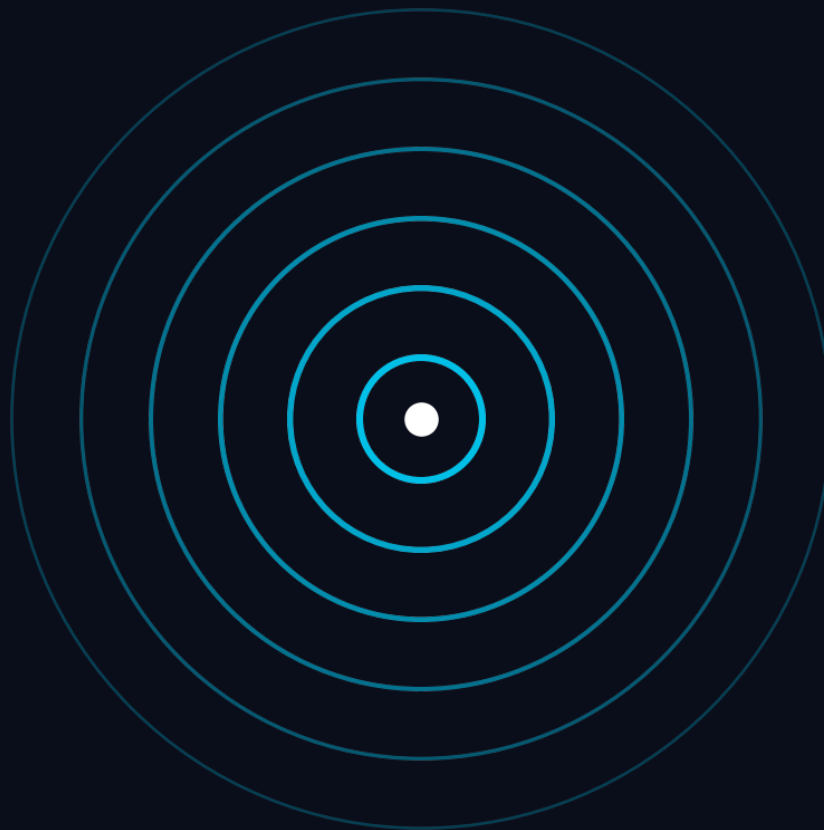
QWEN CLOUD HACKATHON 2026 · ALIBABA
CLOUD

REBOUND

Every phone is a sonar.

Biomimetic echolocation that lets blind users sense walls, doorways, corridors and stairs — with nothing but the phone already in their pocket.

Powered by Qwen-Plus · PyTorch · Web Audio API



Navigation is still analog



43M

people worldwide are blind —
most navigate with a cane or
nothing at all



~1 m

of ground-level reach for a
white cane. Head-height
obstacles and drop-offs go
undetected



\$50K+

and years of waiting for a
guide dog. Electronic aids cost
hundreds and require new
hardware

The one sensor everyone already carries: **a smartphone with a speaker and a microphone.**

What bats knew all along

Bats emit a two-part call — a constant tone to sense motion, a frequency sweep to measure distance — and read the returning echoes to fly through total darkness.

REBOUND does the same with a phone: the speaker emits a biomimetic CF-FM chirp, the microphone captures the room's echo, and the cloud turns that echo into a picture of the space.

No extra hardware. No app store. Just a link.

THE ACTUAL SIGNAL — 20 ms CF-FM CHIRP



8 kHz base: flat MEMS-mic response, audible across all ages, safely under Nyquist at 44.1 kHz.

Chirp to guidance in under a second



EMIT

Speaker fires the 20 ms CF-FM chirp



CAPTURE

Mic records echoes in a latency-aware window



DECONVOLVE

Wiener filter recovers the Room Impulse Response



CLASSIFY

CNN reads 6 space types + DSP stair detector



GUIDE

Qwen agent speaks a personalized instruction

WHAT THE SERVER SEES — STAIRCASE SIGNATURE IN THE RIR

Each step reflects the chirp. Evenly spaced echoes = stairs — detected twice, by the CNN and an independent DSP detector.



A PWA, a GPU-free cloud, and Qwen



MOBILE PWA

- Web Audio chirp + rolling capture
- Gyroscope pitch for stair direction
- Earcons + TTS + haptics
- Zero install — runs from a QR



ALIBABA CLOUD ECS

- FastAPI · Docker · CPU-only
- Wiener deconvolution → RIR
- 6-class CNN, 296K params
- Two-pass DSP stair detector
- Token auth, rate & size limits



QWEN-PLUS

- DashScope API, fully async
- Memory agent per user
- Natural-language guidance

An agent that remembers your world

Qwen-Plus doesn't just caption predictions — it runs the memory. Each observation goes through the agent, which decides what to store, update or deliberately forget.



Episodic memory

Every scan becomes an episode: class, confidence, distance, what the user did next.



Semantic memory

The agent distills recurring patterns — "this user's hallway", "stairs near the kitchen".



Selective forgetting

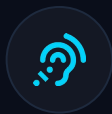
Stale or contradicted memories get their confidence reduced or dropped — decided by the LLM.



Bayesian personalization

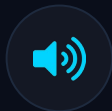
User behavior (advance, hesitate, retreat) reshapes per-class priors across sessions.

Four channels, zero screens needed



Earcons

Rhythm and melodic contour, not just pitch: rising interval = doorway, falling = corner, parking-sensor beeps that accelerate as a wall approaches.



Voice

Built-in TTS speaks "Wall at 1.5 meters" or "Caution: stairs going down" — throttled so it informs without chattering. English and Spanish.



Screen readers

ARIA live regions feed VoiceOver and TalkBack directly — urgent alerts interrupt, ambient updates stay polite.



Haptics

Distinct vibration pattern per space type on Android; a dedicated long alarm for stairs.

Small model, serious rigor

95.3%

validation accuracy across 6 space classes (synthetic RIRs)

0.30 m

mean distance error on validation

296K

CNN parameters — runs on a CPU-only cloud instance

6

classes: open space, wall, doorway, corner, corridor, stairs

88

automated tests passing, including end-to-end pipeline

33

findings from our own red-team audit — all remediated

We attacked our own system first

Before polishing the demo, we ran an adversarial audit against REBOUND and published every finding.



FOUND & FIXED

- Stairs invisible below 27 dB SNR → added stairs as a CNN class + DSP reinforcement
- Path traversal & prompt injection vectors → validation and whitelists everywhere
- Event-loop blocking, race conditions, data leakage in training stats



HARDENED

- Token auth on every endpoint, constant-time compare
- Payload size caps + bounded prediction concurrency (429 over capacity)
- LRU cap on per-user state — memory can't grow unbounded



DISCLOSED

- All metrics measured on simulated rooms — real-world accuracy is an open question we state, not hide
- LIMITATIONS.md ships with the project: empirical values only

LIVE DEMO

Try it on your phone. Right now.

1

Scan the QR — the PWA opens instantly, no install

2

Tap once: microphone + motion permissions, sonar starts

3

Point the phone around the room and listen — beeps, voice, vibration

4

Walk toward the stairwell. REBOUND will warn you first.



SCAN → TAP → SWEEP THE ROOM

WHAT'S NEXT

From demo to daily companion

Real-world dataset

Field RIR collection to close the sim-to-real gap we've documented

On-device personalization

Elastic Weight Consolidation fine-tuning per user (already prototyped)

Edge inference

296K params is small enough to run offline when there's no signal

REBOUND — *because independence shouldn't require special hardware.*